Week 01 teacher guide

Fractions and equal parts

Keep this guide beside PowerPoint slides 1-35. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 001. Saturday | pages 3-14 | slides 4-15
Fractions on one number line

Day 002. Sunday | pages 15-25 | slides 16-35
Fraction operations with models

Day 003. Monday | pages 26-27
Fractions and equal parts

Day 004. Tuesday | pages 28-29
Fractions and equal parts

Day 005. Wednesday | pages 30-31
Fractions and equal parts

Day 006. Thursday | pages 32-33
Fractions and equal parts

Day 007. Friday | pages 34-35
Weekly check

This week’s idea

The whole stays the same when fractions are compared. Multiplication can take part of an amount. Division can count fractional groups or split a fraction into shares.

Use the built-in Booster review where it appears in the day’s work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook’s Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

First-time orientation

Slide 1. Double Advanced Math Program

Say: We will make sense of the math together.

Ask: What would you like to understand better?

Expect: A personal goal or question.

If stuck: Invite a drawing or an example.

Slide 2. How a session runs

Say: The first attempt helps us decide what to discuss.

Ask: Which part gives you time to explain?

Expect: Compare ideas and summarize.

If stuck: Point to the sequence and name each stage.

Slide 3. The twenty sessions

Say: Each week builds on earlier ideas.

Ask: Which topic connects to something you know?

Expect: One named connection.

If stuck: Point to fractions and ratios. These return often.

Saturday. Fractions on one number line

Day 001, pages 3-14 | PowerPoint slides 4-15

Goal: I can compare fractions and decimals by placing them on the same number line.

Set out: paper fraction strips, blank number lines, pencil.

Check first: Name numerator and denominator. Divide a whole into equal parts.

Pacing: Diagnostic 35 + break 5; review 5; lesson 25 (launch 5, explore 8, discuss 10, synthesize 2); exit 5. Total 75 min.

Explore: what to watch for

Uses equal-length wholes. Why it matters: Comparisons require the same unit. Tutor move: Ask the student to mark 0 and 1 on both strips.

Uses a benchmark or common denominator. Why it matters: Both are valid comparison methods. Tutor move: Accept either method and ask where the fractions lie.

Counts spaces between ticks, not tick marks. Why it matters: Each space represents an equal fractional step. Tutor move: Trace one interval and name its length.

Likely misconceptions

Places $\frac{5}{4}$ between 0 and 1. Why: Thinks all fractions are less than one. Tutor move: Build four fourths, then add one fourth.

Orders by denominator size. Why: Confuses number of parts with part size. Tutor move: Compare equal-length strips cut into fifths and ninths.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 4. Week 1, Saturday

Topic: Fractions on one number line
Student: Day 001, pages 3-14

Say: I can compare fractions and decimals by placing them on the same number line.

Ask: What do you already know about this idea?

Expect: Name numerator and denominator. Divide a whole into equal parts.

If stuck: Ask the student to mark 0 and 1 on both strips.

Slide 5. Today | 45 min

Topic: Fractions on one number line
Student: Day 001, pages 3-14

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A fraction names equal parts of one whole. The size of that whole stays fixed.

If stuck: Read the learning goal together.

Slide 6. Launch | 5 min

Topic: Fractions on one number line

Student: Day 001, pages 3-14

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: More pieces always means more muffin. Use the same whole to support your answer.

Expect: P1: $\frac{3}{7} < \frac{5}{7}$

If stuck: Ask the student to mark 0 and 1 on both strips.

Slide 7. Explore | 2 min

Topic: Fractions on one number line

Student: Day 001, pages 3-14

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: More pieces always means more muffin. Use the same whole to support your answer.

Expect: P2: $\frac{4}{5} > \frac{4}{9}$; P3: Less than $\frac{1}{2}$

If stuck: Ask the student to mark 0 and 1 on both strips.

Slide 8. The model | 2 min

Topic: Fractions on one number line

Student: Day 001, pages 3-14

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: One whole keeps the same length.

If stuck: Ask the student to mark 0 and 1 on both strips.

Slide 9. Required practice | 2 min

Topic: Fractions on one number line

Student: Day 001, pages 3-14

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: More pieces always means more muffin. Use the same whole to support your answer.

Expect: P4: $\frac{3}{5} > \frac{7}{12}$; P5: Left to right: 0.5, $\frac{5}{4}$, $1\ \frac{3}{4}$

If stuck: Ask the student to mark 0 and 1 on both strips.

Slide 10. Required practice | 1 min

Topic: Fractions on one number line
Student: Day 001, pages 3-14

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: More pieces always means more muffin. Use the same whole to support your answer.

Expect: P6: $\frac{7}{10} > 0.68$; P7: One fourth of the way from 2 to 4

If stuck: Ask the student to mark 0 and 1 on both strips.

Slide 11. Required practice | 1 min

Topic: Fractions on one number line
Student: Day 001, pages 3-14

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: More pieces always means more muffin. Use the same whole to support your answer.

Expect: P8: $\frac{3}{4}$ of a muffin

If stuck: Ask the student to mark 0 and 1 on both strips.

Slide 12. A claim to check | 10 min

Topic: Fractions on one number line
Student: Day 001, pages 3-14

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: $\frac{4}{5} > \frac{4}{9}$. Nine equal pieces are smaller than five equal pieces of the same whole.

If stuck: Align equal-length strips. Cover four fifths and four ninths.

Slide 13. In your own words | 2 min

Topic: Fractions on one number line
Student: Day 001, pages 3-14

Say: Use your own words and one example.

Ask: Agree or disagree: More pieces always means more muffin. Use the same whole to support your answer.

Expect: A fraction names equal parts of one whole. The size of that whole stays fixed.

If stuck: Point to the model and name what each part represents.

Slide 14. Exit ticket | 5 min

Topic: Fractions on one number line

Student: Day 001, pages 3-14

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: $\frac{7}{10}$, 0.72, $\frac{3}{4}$

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 15. Tutor reference

Topic: Fractions on one number line

Student: Day 001, pages 3-14

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: $\frac{7}{10}$, 0.72, $\frac{3}{4}$

Got it: Orders all three correctly and places them in order between 0.7 and 0.8, with $\frac{3}{4}$ at 0.75.

Almost there: Order is correct, but a position or label is imprecise; explains that larger denominator means smaller equal parts.

Needs re-teaching: Reverses the fractions because 10 exceeds 4, or uses inconsistent units.

Sunday. Fraction operations with models

Day 002, pages 15-25 | PowerPoint slides 16-35

Goal: I can use equal parts to explain fraction multiplication and two kinds of division.

Set out: paper rectangles, fraction strips, pencil.

Check first: Equivalent fractions and equal-sized parts.

Pacing: Retrieval 5; launch 5; explore 15; discuss 12; synthesis 5; mixed review 10; exit 8; preview 5. Total 65 min.

Explore: what to watch for

Counts overlapping equal parts in the area model. Why it matters: The overlap represents multiplication. Tutor move: Ask what fraction of the entire unit square is shaded twice.

Predicts shrinking for a factor below one. Why it matters: Connects computation to scaling. Tutor move: Ask for an estimate before calculation.

Distinguishes number of pieces from piece size. Why it matters: These division situations have different units. Tutor move: Accept a tape or a story, but require units in the answer.

Likely misconceptions

Says multiplication always increases an amount. Why: Uses only whole-number experience. Tutor move: Take half of a familiar amount.

Uses the same answer for reversed division. Why: Does not identify what is known and unknown. Tutor move: Label total amount, number of groups, and group size.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 16. Week 1, Sunday

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: I can use equal parts to explain fraction multiplication and two kinds of division.

Ask: What do you already know about this idea?

Expect: Equivalent fractions and equal-sized parts.

If stuck: Ask what fraction of the entire unit square is shaded twice.

Slide 17. Today

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Multiplication can take part of an amount. Division can count groups or find the size of each share.

If stuck: Read the learning goal together.

Slide 18. Recall | 3 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 56; 2: 8; 3: $\frac{3}{4}$; 4: 0.85

If stuck: Give time to draw before giving a hint.

Slide 19. Recall: answers | 2 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 56; 8; $\frac{3}{4}$; 0.85

If stuck: Rebuild one missed item with its model.

Slide 20. Launch | 5 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: Taking three fourths of a cup gives more than one cup.

Expect: P1: 3 $\frac{3}{4}$ cups

If stuck: Ask what fraction of the entire unit square is shaded twice.

Slide 21. Explore | 3 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: Taking three fourths of a cup gives more than one cup.

Expect: P2: $\frac{2}{5}$ square meter

If stuck: Ask what fraction of the entire unit square is shaded twice.

Slide 22. The model | 2 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: $\frac{2}{3}$ of $\frac{3}{5}$ is $\frac{6}{15}$, or $\frac{2}{5}$.

If stuck: Ask what fraction of the entire unit square is shaded twice.

Slide 23. Required practice | 2 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: Taking three fourths of a cup gives more than one cup.

Expect: P3: 1 cup; P4: $6 < 8$; $10 > 8$

If stuck: Ask what fraction of the entire unit square is shaded twice.

Slide 24. Required practice | 2 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: Taking three fourths of a cup gives more than one cup.

Expect: P5: 12 pieces; P6: $\frac{1}{12}$ liter

If stuck: Ask what fraction of the entire unit square is shaded twice.

Slide 25. Required practice | 2 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: Taking three fourths of a cup gives more than one cup.

Expect: P7: 10 groups; $\frac{1}{10}$ in each share. Stories vary.

If stuck: Ask what fraction of the entire unit square is shaded twice.

Slide 26. Watch the thinking | 2 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: $\frac{1}{2}$ cup

If stuck: Return to the matching unit in the model.

Slide 27. Your turn | 2 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$

If stuck: Compare with the previous example without copying its numbers.

Slide 28. Compare methods | 6 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Partner compare: show the same problem in two ways.

Ask: Agree or disagree: Taking three fourths of a cup gives more than one cup. Partner compare: In the two ribbon stories, which question counts pieces and which finds their size?

Expect: Accept drawings and equations that preserve the same quantities and units. Predicts shrinking for a factor below one.

If stuck: Ask for an estimate before calculation.

Slide 29. A claim to check | 6 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: Twelve quarter-meter pieces fit in 3 meters.

If stuck: Make one meter from four quarter-meter pieces, then repeat three times.

Slide 30. Words and meaning | 3 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: $\frac{1}{2} \times 8$ takes half of eight. $\frac{1}{5}$ is one fifth. $\frac{1}{2} \times 8$ has product 4.

If stuck: Read one definition and draw its example.

Slide 31. Mixed review | 10 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Choose a method for each problem.

Ask: What tells you which operation to use?

Expect: 52; 8; $\frac{1}{2}$

If stuck: Name the quantities and units before calculating.

Slide 32. In your own words | 2 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Use your own words and one example.

Ask: Agree or disagree: Taking three fourths of a cup gives more than one cup.

Expect: Multiplication can take part of an amount. Division can count groups or find the size of each share.

If stuck: Point to the model and name what each part represents.

Slide 33. Exit ticket | 8 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 10 shelves

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 34. The week ahead | 5 min

Topic: Fraction operations with models
Student: Day 002, pages 15-25

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 35. Tutor reference

Topic: Fraction operations with models

Student: Day 002, pages 15-25

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 10 shelves

Got it: Gives 10 shelves and connects five meters to ten equal half-meter pieces.

Almost there: Gives 10 but cannot yet explain the unit or draw equal pieces.

Needs re-teaching: Gives 2 $\frac{1}{2}$ or claims the answer must be below 5.

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 003. Monday | pages 26-27

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Fractions: compare and combine: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 004. Tuesday | pages 28-29

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Fraction operations: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 005. Wednesday | pages 30-31

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 006. Thursday | pages 32-33

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Multiplication and division facts: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 007. Friday | pages 34-35

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Diagnostic: fact snapshot

Day 001, student workbook page 3

Diagnostic scoring: record each skill separately. Treat 80% or more correct as secure, 50-79% as developing, and below 50% as a priority for review. A single sampled item gives provisional evidence. Keep the fact snapshot separate from reasoning; unfamiliar content is not a failure.

1. 5 × 4 = 20	2. 3 × 6 = 18	3. 7 × 11 = 77
4. 7 × 8 = 56	5. 8 × 8 = 64	6. 4 × 3 = 12
7. 7 × 6 = 42	8. 5 × 7 = 35	9. 8 × 11 = 88
10. 2 × 11 = 22	11. 7 × 9 = 63	12. 6 × 9 = 54
13. 8 × 2 = 16	14. 12 × 10 = 120	15. 4 × 5 = 20
16. 7 × 4 = 28	17. 3 × 5 = 15	18. 2 × 4 = 8
19. 4 × 12 = 48	20. 6 × 10 = 60	21. 11 × 2 = 22
22. 2 × 5 = 10	23. 6 × 11 = 66	24. 6 × 3 = 18
25. 8 × 7 = 56	26. 6 × 12 = 72	27. 7 × 10 = 70
28. 8 × 6 = 48	29. 7 × 2 = 14	30. 12 × 7 = 84

Diagnostic: computation and reasoning

Day 001, student workbook pages 4-8

Diagnostic scoring: record each skill separately. Treat 80% or more correct as secure, 50-79% as developing, and below 50% as a priority for review. A single sampled item gives provisional evidence. Keep the fact snapshot separate from reasoning; unfamiliar content is not a failure.

1 [N-MULT] | student p. 4

Answer: 9798

2 [N-DIV] | student p. 4

Answer: 228

3 [N-DIV] | student p. 4

Answer: 91

4 [D-OPS] | student p. 4

Answer: 18.45

5 [D-OPS] | student p. 4

Answer: 4.63

6 [D-OPS] | student p. 4

Answer: 0.28

7 [D-OPS] | student p. 5

Answer: 7

8 [F-MULT] | student p. 5

Answer: $\frac{1}{2}$

9 [F-ADD] | student p. 5

Answer: $1\ \frac{1}{12}$

10 [F-DIV] | student p. 5

Answer: 12

11 [F-DIV] | student p. 5

Answer: $\frac{1}{15}$

12 [F-MULT] | student p. 5

Answer: 7

13 [F-COMP] | student p. 6

Answer: $\frac{4}{9} < \frac{1}{2}$

14 [F-COMP] | student p. 6

Answer: Left to right: $\frac{5}{8}$, $\frac{3}{4}$, $\frac{4}{3}$, $1\ \frac{1}{2}$

15 [D-COMP] | student p. 6

Answer: 0.506, 0.56, 0.6

16 [R-RATIO] | student p. 6

Answer: 10

Reasoning: 15 reds form five groups. Each group has 2 blues.

17 [R-PERCENT] | student p. 6

Answer: 16

18 [E-EXPR] | student p. 6

Answer: 27

19 [N-SIGNED] | student p. 7

Answer: 6°C

20 [G-AREA] | student p. 7

Answer: $\frac{1}{2}$ m²

21 [G-VOL] | student p. 7

Answer: 60 cm³

22 [N-COORD] | student p. 7

Answer: 3 units

23 [F-SCALE] | student p. 7

Answer: 6 is two of three equal shares of 9. A factor between 0 and 1 scales down.

24 [P-LOGIC] | student p. 7

Answer: 20, 27

Reasoning: The jumps are 3, 4, 5, then 6 and 7.

25 [P-LOGIC] | student p. 8

Answer: 4, 6, 8

Reasoning: Remove the extra 2 and 4. Divide the remaining 12 equally.

Fractions on one number line

Day 001, student workbook pages 9-13

P1 [F-COMP] | student p. 9

Answer: $\frac{3}{7} < \frac{5}{7}$

Reasoning: Both use sevenths. Three parts are fewer than five.

Watch for >: Counting denominator size instead of the number of parts.

P2 [F-COMP] | student p. 9

Answer: $\frac{4}{5} > \frac{4}{9}$

Reasoning: Fifths are larger than ninths, so four fifths cover more.

Watch for <: Thinking a larger denominator makes a larger amount.

P3 [F-COMP] | student p. 9

Answer: Less than $\frac{1}{2}$

Reasoning: Half of twelve parts is six parts. Five twelfths is below half.

P4 [F-COMP] | student p. 10

Answer: $\frac{3}{5} > \frac{7}{12}$

Reasoning: $\frac{3}{5} = \frac{36}{60}$; $\frac{7}{12} = \frac{35}{60}$. Compare 36 and 35.

P5 [F-LINE] | student p. 10

Answer: Left to right: 0.5, $\frac{5}{4}$, $1\ \frac{3}{4}$

Reasoning: $\frac{5}{4} = 1\ \frac{1}{4}$. Split each whole into four equal steps.

P6 [F-COMP] | student p. 10

Answer: $\frac{7}{10} > 0.68$

Reasoning: $\frac{7}{10} = 0.70$, which is 2 hundredths more than 0.68.

P7 [F-LINE] | student p. 11

Answer: One fourth of the way from 2 to 4

Reasoning: The interval from 2 to 4 is two units. Half a unit uses one fourth of that interval.

P8 [F-QUOT] | student p. 11

Answer: $\frac{3}{4}$ of a muffin

Reasoning: Split each muffin into fourths. Each friend gets three fourths.

F [F-COMP] | student p. 11

Answer: 10 twelfths; $\frac{5}{6} > \frac{3}{4}$

Reasoning: Use the worked example in the slides to check each step.

D [F-COMP] | student p. 12

Answer: $\frac{4}{5} > \frac{4}{9}$. Nine equal pieces are smaller than five equal pieces of the same whole.

Reasoning: She compares denominator counts without comparing piece sizes. Align equal-length strips. Cover four fifths and four ninths.

S [EXPLAIN] | student p. 12

Answer: Answers vary. A fraction names equal parts of one whole. The size of that whole stays fixed.

Reasoning: Accept an accurate explanation supported by a matching example.

E [F-COMP] | student p. 13

Answer: $\frac{7}{10}$, 0.72, $\frac{3}{4}$

Reasoning: Rename as 0.70, 0.72, 0.75. Their positions increase to the right.

Watch for $\frac{3}{4}$, 0.72, $\frac{7}{10}$: The larger denominator does not make the fraction larger.

Fraction operations with models

Day 002, student workbook pages 15-20

R1 [G5-FACT] | student p. 15

Answer: 56

R2 [G5-FACT] | student p. 15

Answer: 8

R3 [F-ADD] | student p. 15

Answer: $\frac{3}{4}$

R4 [D-ADD] | student p. 15

Answer: 0.85

S [EXPLAIN] | student p. 15

Answer: Answers vary. Multiplication can take part of an amount. Division can count groups or find the size of each share.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [F-MULT] | student p. 16

Answer: $3\ \frac{3}{4}$ cups

Reasoning: Five groups of three fourths make fifteen fourths.

P2 [F-MULT] | student p. 16

Answer: $\frac{2}{5}$ square meter

Reasoning: Divide a unit square into 5 columns and 3 rows. The overlap is 6 of 15 parts.

P3 [F-MULT] | student p. 16

Answer: 1 cup

Reasoning: Rename $1\ \frac{1}{2}$ as $\frac{3}{2}$. Two thirds of three halves is two halves.

P4 [F-SCALE] | student p. 17

Answer: $6 < 8$; $10 > 8$

Reasoning: Three fourths takes less than all eight. Five fourths takes more than all eight.

P5 [F-DIV] | student p. 17

Answer: 12 pieces

Reasoning: There are four fourths per meter. Three meters hold twelve fourths.

P6 [F-DIV] | student p. 17

Answer: $\frac{1}{12}$ liter

Reasoning: Split one fourth into three equal parts. Each is one twelfth of a liter.

P7 [F-DIV] | student p. 18

Answer: 10 groups; $\frac{1}{10}$ in each share. Stories vary.

Reasoning: Example: cut 2 meters into fifth-meter pieces; share a fifth-meter piece equally between two people.

F [F-DIV] | student p. 18

Answer: $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$

Reasoning: Use the worked example in the slides to check each step.

D [F-DIV] | student p. 18

Answer: Twelve quarter-meter pieces fit in 3 meters.

Reasoning: He treats division by a fraction as multiplication by that fraction. Make one meter from four quarter-meter pieces, then repeat three times.

M1 [G5-FACT] | student p. 19

Answer: 52

M2 [G5-FACT] | student p. 19

Answer: 8

M3 [F-ADD] | student p. 19

Answer: $\frac{1}{2}$

E [F-DIV] | student p. 20

Answer: 10 shelves

Reasoning: Each meter contains two half-meters, so five meters contain ten.

Watch for 2 $\frac{1}{2}$: Multiplying by one half answers a different question.

Study guide: Try it

Day 002, student workbook pages 21-24

1 [F-COMP] | student p. 24

Answer: $\frac{2}{9} < \frac{7}{9}$

Reasoning: With ninths fixed, compare two parts and seven parts.

Watch for $\frac{2}{9} > \frac{7}{9}$: The numerator counts the parts.

2 [F-COMP] | student p. 24

Answer: $\frac{7}{8}$

Reasoning: The gaps from 1 are one eighth and one fifth. One eighth is smaller.

3 [F-MULT] | student p. 24

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

4 [F-MULT] | student p. 24

Answer: 1 cup

Reasoning: Rename one and two thirds as five thirds. Three fifths of five thirds is one.

5 [F-SCALE] | student p. 24

Answer: Smaller; 6

Reasoning: Two thirds takes less than all nine.

Daily practice

Day 003, student workbook pages 26-27

1 [N-FACT] | student p. 26

Answer: 9

2 [N-FACT] | student p. 26

Answer: 21

3 [N-FACT] | student p. 26

Answer: 12

4 [N-FACT] | student p. 26

Answer: 27

5 [F-COMP] | student p. 26

Answer: $\frac{1}{3} > \frac{1}{4}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{1}{3} < \frac{1}{4}$: A larger denominator makes smaller equal parts, not a larger fraction.

6 [F-ADD] | student p. 26

Answer: $\frac{5}{6}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

7 [F-ADD] | student p. 26

Answer: $\frac{11}{12}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

8 [F-SUB] | student p. 26

Answer: $\frac{5}{12}$

Reasoning: Rename both fractions with a common denominator. Subtract equal parts.

9 [F-COMP] | student p. 27 | Built-in review: Fractions: compare and combine

Answer: $\frac{2}{9} < \frac{7}{9}$

Reasoning: With ninths fixed, compare two parts and seven parts.

Watch for $\frac{2}{9} > \frac{7}{9}$: The numerator counts the parts.

10 [F-COMP] | student p. 27 | Built-in review: Fractions: compare and combine

Answer: $\frac{2}{4} > \frac{2}{5}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{2}{4} < \frac{2}{5}$: A larger denominator makes smaller equal parts, not a larger fraction.

11 [F-COMP] | student p. 27 | Built-in review: Fractions: compare and combine

Answer: $\frac{2}{5}$, 0.45, $\frac{1}{2}$

Reasoning: Rename two fifths as 0.40 and one half as 0.50.

12 [F-COMP] | student p. 27 | Built-in review: Fractions: compare and combine

Answer: $\frac{5}{8} > \frac{3}{5}$

Reasoning: Rename as twenty-five fortieths and twenty-four fortieths.

Watch for $\frac{5}{8} < \frac{3}{5}$: Compare equal-sized parts, not the original denominator counts.

13 [F-COMP] | student p. 27

Answer: $\frac{4}{5} > \frac{4}{9}$. Nine equal pieces are smaller than five equal pieces of the same whole.

Watch for $\frac{4}{9} > \frac{4}{5}$: She compares denominator counts without comparing piece sizes.

Daily practice

Day 004, student workbook pages 28-29

1 [F-EQUIV] | student p. 28

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 28

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 28

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 28

Answer: 0.125; 12.5%

5 [F-DIV] | student p. 28

Answer: 20; $\frac{1}{20}$

Reasoning: The first counts fifths in four. The second splits one fifth into four shares.

Watch for $\frac{4}{5}$; $\frac{4}{5}$: Changing the order changes what is shared or counted.

6 [F-MULT] | student p. 28

Answer: $\frac{2}{3}$ square meter

Reasoning: Multiply the two side lengths and simplify twenty thirtieths.

7 [F-SCALE] | student p. 28

Answer: Smaller; 6

Reasoning: Two thirds takes less than all nine.

8 [N-FACT] | student p. 28

Answer: 15

9 [F-MULT] | student p. 29 | Built-in review: Fraction operations

Answer: 2

Reasoning: Count two thirds for each whole-number group, then rename the result.

10 [F-MULT] | student p. 29 | Built-in review: Fraction operations

Answer: $\frac{4}{15}$

11 [F-DIV] | student p. 29 | Built-in review: Fraction operations

Answer: 6 pieces

Reasoning: Each meter contains 3 equal pieces. Multiply by 2 meters.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

12 [F-DIV] | student p. 29 | Built-in review: Fraction operations

Answer: $\frac{1}{6}$ liter

Reasoning: Split the given fractional amount into the stated number of equal shares.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

13 [F-DIV] | student p. 29

Answer: Twelve quarter-meter pieces fit in 3 meters.

Watch for $\frac{3}{4}$: He treats division by a fraction as multiplication by that fraction.

Daily practice

Day 005, student workbook pages 30-31

1 [N-SCALE] | student p. 30

Answer: 32

2 [N-SCALE] | student p. 30

Answer: 3

3 [N-SCALE] | student p. 30

Answer: 7

4 [N-SCALE] | student p. 30

Answer: 9

5 [F-LINE] | student p. 30

Answer: At 1 $\frac{3}{4}$, the seventh quarter-step from 0

Reasoning: Four fourths reach 1. Three more fourths reach one and three fourths.

6 [F-DIV] | student p. 30

Answer: $\frac{1}{18}$; stories vary

Reasoning: Example: share one sixth liter equally among three cups. Each gets one eighteenth liter.

7 [N-FACT] | student p. 30

Answer: 24

8 [F-ADD] | student p. 30

Answer: $\frac{1}{3}$

9 [N-SCALE] | student p. 31

Answer: 3

10 [N-FACT] | student p. 31

Answer: 6

11 [N-FACT] | student p. 31

Answer: 36

12 [G-VOL] | student p. 31

Answer: 108 cm³

13 [PUZZLE] | student p. 31 | Built-in review: Weekly puzzle

Answer: $\frac{5}{8}$

Reasoning: Half is four eighths and three fourths is six eighths. Only five eighths lies strictly between them.

Daily practice

Day 006, student workbook pages 32-33

1 [N-EST] | student p. 32

Answer: 300

2 [N-EST] | student p. 32

Answer: 4

3 [N-EST] | student p. 32

Answer: About 100

4 [N-EST] | student p. 32

Answer: About 20

5 [F-LINE] | student p. 32

Answer: At 1 $\frac{3}{4}$, the seventh quarter-step from 0

Reasoning: Four fourths reach 1. Three more fourths reach one and three fourths.

6 [F-DIV] | student p. 32

Answer: $\frac{1}{18}$; stories vary

Reasoning: Example: share one sixth liter equally among three cups. Each gets one eighteenth liter.

7 [F-COMP] | student p. 32

Answer: $\frac{5}{8} > \frac{3}{5}$

Reasoning: Rename as twenty-five fortieths and twenty-four fortieths.

Watch for $\frac{5}{8} < \frac{3}{5}$: Compare equal-sized parts, not the original denominator counts.

8 [G-VOL] | student p. 32

Answer: 30 cm³

9 [N-FACT] | student p. 33 | Built-in review: Multiplication and division facts

Answer: 48

10 [N-FACT] | student p. 33 | Built-in review: Multiplication and division facts

Answer: 8

11 [N-FACT] | student p. 33 | Built-in review: Multiplication and division facts

Answer: 63

12 [N-FACT] | student p. 33 | Built-in review: Multiplication and division facts

Answer: 9

13 [F-SCALE] | student p. 33

Answer: Smaller; 6

Reasoning: Two thirds takes less than all nine.

14 [F-DIV] | student p. 33

Answer: Twelve quarter-meter pieces fit in 3 meters.

Watch for $\frac{3}{4}$: He treats division by a fraction as multiplication by that fraction.

Weekly check

Day 007, student workbook pages 34-35

1 [F-LINE] | student p. 34

Answer: At 1 $\frac{3}{4}$, the seventh quarter-step from 0

Reasoning: Four fourths reach 1. Three more fourths reach one and three fourths.

2 [F-DIV] | student p. 34

Answer: $\frac{1}{18}$; stories vary

Reasoning: Example: share one sixth liter equally among three cups. Each gets one eighteenth liter.

3 [F-COMP] | student p. 34

Answer: $\frac{5}{8} > \frac{3}{5}$

Reasoning: Rename as twenty-five fortieths and twenty-four fortieths.

Watch for $\frac{5}{8} < \frac{3}{5}$: Compare equal-sized parts, not the original denominator counts.

4 [F-ADD] | student p. 34

Answer: $\frac{5}{6}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

5 [G-VOL] | student p. 34

Answer: 54 cm³

6 [N-FACT] | student p. 35

Answer: 12

7 [D-OPS] | student p. 35

Answer: 6.75

8 [N-SCALE] | student p. 35

Answer: 5

9 [N-FACT] | student p. 35

Answer: 20

10 [F-COMP] | student p. 35

Answer: $\frac{4}{5} > \frac{4}{9}$. Nine equal pieces are smaller than five equal pieces of the same whole.

Watch for $\frac{4}{9} > \frac{4}{5}$: She compares denominator counts without comparing piece sizes.